1. (a)
$$p(p|x) \propto p^{a-1}(1-p)^{b-1}(1-p)^{k-1}p = p^a(1-p)^{b-k-2},$$
which is the kernel of the Beta$(a+1, b-k-1)$-distribution. Hence, this is a conjugate pair.

   (b)
$$p(\mu|x) \propto \frac{\mathbb{1}_{[-1,1]}(\mu)}{2}\frac{1}{\sqrt{2\pi}}e^{-(x-\mu)^2/2}$$
is not a uniform distribution. Hence, this is not a conjugate pair.

   (c)
$$p(\beta|x) \propto \frac{b^a}{\Gamma(a)}\beta^{a-1}e^{-b\beta}\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x} \propto \beta^{a+\alpha-1}e^{-(b+x)\beta},$$
which is the kernel of the Gamma$(a+\alpha, b+x)$-distribution. Hence, this is a conjugate pair.

2. Let $X = (X_1, \ldots, X_n)|\lambda \sim \text{Pois}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$. The posterior is proportional to
$$p(\lambda|X) \propto \lambda^{a-1}e^{-b\lambda}\prod_{i=1}^n e^{-\lambda}\lambda^{x_i} = e^{-(b+n)\lambda}\lambda^{a-1+\sum_{i=1}^n x_i}.$$
The right-hand side is the kernel of a Gamma$(a+\sum_{i=1}^n x_i, b+n)$-distribution, which verifies conjugacy. The mean of a Gamma(a,b) distribution is a/b, so that the posterior mean is
$$\frac{a+\sum_{i=1}^n x_i}{b+n} = \frac{b}{b+n}\frac{a}{b} + \frac{n}{b+n}\bar{x}.$$
As the sample size n increases, the posterior mean shifts from the prior mean to the sample mean.

3. Given i.i.d data $X = (X_1, \ldots, X_n) \sim \text{Exp}(\lambda)$, the log-likelihood and its second derivative are
$$\begin{aligned}\ell(\lambda|X) &= n\log(\lambda) - \lambda n\bar{X},\\ \ell''(\lambda|X) &= -n/\lambda^2.\end{aligned}$$
Since $\ell''(\lambda|X)$ is independent of X, the Jeffrey's prior density is
$$p(\lambda) \propto \sqrt{\mathbb{E}_\lambda\left[\frac{n}{\lambda^2}\right]} = \frac{\sqrt{n}}{\lambda}$$
on $\lambda > 0$. Its integral does not converge so the prior is improper. But the posterior is
$$\begin{aligned}p(\lambda|X) &\propto p(\lambda)L(\lambda|X) = \frac{\sqrt{n}}{\lambda}\lambda^n e^{-\lambda n\bar{X}}\\ &= \sqrt{n}\lambda^{n-1}e^{-\lambda n\bar{X}}.\end{aligned}$$
This is the kernel of a Gamma$(n, n\bar{X})$-distribution, and hence the posterior is proper. Alternatively, one can check directly that the integral of $p(\lambda|X)$ over $\lambda \in [0, \infty)$ converges.

4. From lecture notes we know that the prior and likelihood are conjugate, and that the posterior is

$$\mu|\mathbf{x} \sim N\left(\frac{1}{n/\sigma^2 + 1/s^2}\left(\frac{m}{s^2} + \frac{n\bar{\mathbf{x}}}{\sigma^2}\right), \frac{1}{n/\sigma^2 + 1/s^2}\right).$$

For brevity, let

$$a := \frac{1}{n/\sigma^2 + 1/s^2}\left(\frac{m}{s^2} + \frac{n\bar{\mathbf{x}}}{\sigma^2}\right),$$
$$b := \sqrt{\frac{1}{n/\sigma^2 + 1/s^2}},$$

so that the posterior is the $N(a, b^2)$-distribution. Via another completion of the square, the posterior predictive density of a new observation X_{n+1} can be written as

$$\begin{aligned}
&p(x|\mathbf{x}) \\
&= \int_{-\infty}^{\infty} p(x|\mu)p(\mu|\mathbf{x})\mathrm{d}\mu \propto \int_{-\infty}^{\infty} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)\exp\left(\frac{-(\mu-a)^2}{2b^2}\right)\mathrm{d}\mu \\
&\propto \int_{-\infty}^{\infty} \exp\left(\frac{-1}{2}\left\{\left(\frac{1}{\sigma^2} + \frac{1}{b^2}\right)\mu^2 - 2\left(\frac{x}{\sigma^2} + \frac{a}{b^2}\right)\mu + \frac{x^2}{\sigma^2}\right\}\right)\mathrm{d}\mu \\
&= \exp\left(\frac{-x^2}{2\sigma^2}\right)\int_{-\infty}^{\infty} \exp\left(\frac{-1}{2}\left(\frac{1}{\sigma^2} + \frac{1}{b^2}\right)\left\{\mu^2 - 2\left(\frac{1}{\sigma^2} + \frac{1}{b^2}\right)^{-1}\left(\frac{x}{\sigma^2} + \frac{a}{b^2}\right)\mu\right\}\right)\mathrm{d}\mu \\
&= \exp\left(\frac{-x^2}{2\sigma^2}\right)\int_{-\infty}^{\infty} \exp\left(\frac{-1}{2}\left(\frac{1}{\sigma^2} + \frac{1}{b^2}\right)\left\{\left(\mu - \frac{b^2x + \sigma^2 a}{b^2 + \sigma^2}\right)^2 - \frac{(xb^2 + a\sigma^2)^2}{(b^2+\sigma^2)^2}\right\}\right)\mathrm{d}\mu \\
&\propto \exp\left(\frac{-x^2}{2\sigma^2}\right)\exp\left(\frac{1}{2}\left(\frac{1}{\sigma^2} + \frac{1}{b^2}\right)\frac{(xb^2 + a\sigma^2)^2}{(b^2+\sigma^2)^2}\right) \\
&\propto \exp\left(\frac{1}{2\sigma^2}\left(\frac{x^2b^4 + a^2\sigma^4 + 2a\sigma^2b^2x}{b^2(b^2+\sigma^2)} - x^2\right)\right) \\
&\propto \exp\left(\frac{-(x-a)^2}{2(b^2+\sigma^2)}\right).
\end{aligned}$$

Throughout the calculation, the missing normalising constants in the lines beginning with $\propto$ absorb terms which don't depend on either x or μ.

Substituting back the values of a and b, the posterior predictive distribution is

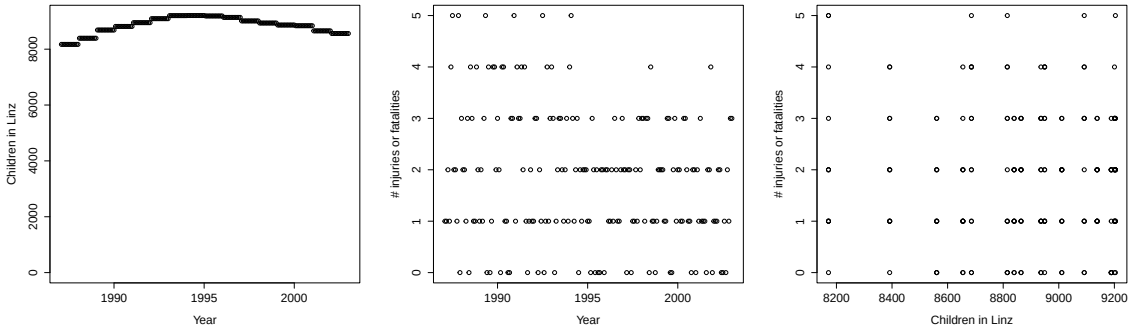
$$X_{n+1}|\mathbf{x} \sim N\left(\frac{1}{n/\sigma^2 + 1/s^2}\left(\frac{m}{s^2} + \frac{n\bar{\mathbf{x}}}{\sigma^2}\right), \frac{1}{n/\sigma^2 + 1/s^2} + \sigma^2\right).$$

5. (a) The following code visualises the data:

```
load("children.Rdata")
```

```
plot(1987 + (1:nrow(children))/12, children$exposed, xlab="
   Year", ylab="Children in Linz", ylim=c(0,max(children$
   exposed)))
plot(1987 + (1:nrow(children))/12, children$y, xlab="Year",
   ylab="# injuries or fatalities")
plot(children$exposed, children$y, xlab="Children in Linz",
   ylab="# injuries or fatalities")
```

The output looks like this:



The number of children does not vary much, the number of injuries and fatalities is low, and there is no obvious relationship between the variables.

(b) A reasonable model is

$$Y_t \sim \text{Bin}(n_t, \pi),$$

where Y_t is the number of injuries or fatalities in month t, n_t is the number of children in that month, and all random variables are independent. This yields the likelihood

$$L(\pi|y_1, \ldots, y_m) = \prod_{t=1}^{m} \binom{n_t}{y_t} \pi^{y_t}(1-\pi)^{n_t - y_t},$$

which is maximised by

$$\hat{\pi}_{\text{MLE}} = \frac{\sum_{t=1}^{m} y_t}{\sum_{t=1}^{m} n_t}.$$

The Fisher information is

$$-\ell''(\pi|y_1, \ldots, y_m) = \frac{\sum_{t=1}^{m} y_t}{\pi^2} + \frac{\sum_{t=1}^{m}(n_t - y_t)}{(1-\pi)^2},$$

so that the observed Fisher information cancels down to

$$-\ell''(\hat{\pi}_{\text{MLE}}|y_1, \ldots, y_m) = \frac{(\sum_{t=1}^{m} n_t)^3}{(\sum_{t=1}^{m} y_t)(\sum_{t=1}^{m}\{n_t - y_t\})}.$$

Hence, an approximate 95% confidence interval is

$$\left[\frac{\sum_{t=1}^{m} y_t}{\sum_{t=1}^{m} n_t} \pm z_{0.975}\frac{\sqrt{\sum_{t=1}^{m} y_t}\sqrt{\sum_{t=1}^{m}(n_t - y_t)}}{\sqrt{(\sum_{t=1}^{m} n_t)^3}}\right],$$

where z_a is the a-quantile of the standard Gaussian distribution.

6. (a) The posterior density is proportional to

$$p(\mu|y_1,\dots,y_m) \propto e^{\mu}\prod_{t=1}^{m} e^{-\mu}\mu^{y_t} = e^{-(m+1)\mu}\mu^{\sum_{t=1}^m y_t},$$

i.e. $\mu|y_1,\dots y_t \sim \text{Gamma}(\sum_{t=1}^m y_t + 1, m+1)$. Reasonable point estimates include the posterior mean or mode:

$$\hat{\mu}_2 = \frac{\sum_{t=1}^m y_t + 1}{m+1} \approx 1.834,$$
$$\hat{\mu}_\infty = \frac{\sum_{t=1}^m y_t}{m+1} \approx 1.829.$$

The median for a Gamma distribution is not available in closed form. A natural credible interval is given by the 0.025 and 0.975 quantiles of the posterior. Since $m = 192$ and $\sum_{t=1}^m y_t = 353$, we obtain the interval

$$\Big[\text{qgamma}(0.025, \sum_{t=1}^m y_t + 1, m+1), \text{qgamma}(0.975, \sum_{t=1}^m y_t + 1, m+1)\Big]$$
$$\approx [1.648, 2.030].$$

(b) Varying the prior hyperparameter only changes the rate of the posterior Gamma distribution. E.g. with $\mu \sim \text{Exp}(0.1)$ the posterior is $\mu|y_1,\dots y_t \sim \text{Gamma}(\sum_{t=1}^m y_t + 1, m + 0.1)$, with posterior mean and credible interval

$$\hat{\mu}_2 = \frac{\sum_{t=1}^m y_t + 1}{m+1} \approx 1.843, \qquad [1.656, 2.040],$$

while with prior $\mu \sim \text{Exp}(10)$ we have

$$\hat{\mu}_2 = \frac{\sum_{t=1}^m y_t + 1}{m+10} \approx 1.752, \qquad [1.575, 1.940].$$

Clearly the posterior is not overly sensitive to the choice of prior, which is desirable for a reasonable sample size.

7. By the law of large numbers, $\bar{\mathbf{X}} \to \mu$ in probability (and almost surely). Thus

$$\lim_{n\to\infty} \mathbb{E}_{\mu_0}[\mathbb{1}\{|\bar{\mathbf{X}} - \mu_0| > \varepsilon\}] = \lim_{n\to\infty} \mathbb{P}_{\mu_0}(|\bar{\mathbf{X}} - \mu_0| > \varepsilon)$$
$$= \mathbb{1}\{|\mu_0 - \mu_0| > \varepsilon\} = 0.$$

Similarly, fix $\mu_1 : |\mu_1 - \mu_0| > \varepsilon$. Then

$$\lim_{n\to\infty} \sup_{\mu:\|\mu-\mu_0\|>\varepsilon} \{\mathbb{E}_{\mu}[\mathbb{1}\{|\bar{\mathbf{X}} - \mu_0| > \varepsilon\}]\} \geq \lim_{n\to\infty} \mathbb{P}_{\mu_1}(|\bar{\mathbf{X}} - \mu_0| > \varepsilon)$$
$$= \mathbb{1}\{|\mu_1 - \mu_0| > \varepsilon\} = 1.$$

As an aside, note that this solution is not specific to the Gaussian in any way. The same argument would work for estimating the mean of essentially any distribution using i.i.d. data, as long as the mean is finite and the law of large numbers holds.